

Databases – Exercises Database Normalisation

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1. We consider a relation $R(A, B, C, D, E, F, G)$ with functional dependencies:

$$\begin{array}{llll} A \rightarrow E, F & A, B, C \rightarrow D, G & E \rightarrow C, G & C \rightarrow B, G \\ B, C \rightarrow E & A, B, C \rightarrow D & F \rightarrow D, E & E, G \rightarrow C, D \end{array}$$

- (a) Compute the cover of $\{C\}$.
(b) Determine which of the functional dependencies

$$C \rightarrow A, F \quad C \rightarrow D \quad C \rightarrow B, D \quad C \rightarrow F \quad C \rightarrow A, C$$

is implied by the functional dependencies above.

2. We consider a relation $R(A, B, C, D, E)$ with functional dependencies:

$$A \rightarrow B \quad A, D, E \rightarrow B, C \quad B, E \rightarrow A \quad C \rightarrow B, E$$

Compute all minimal keys of the relation R .

3. We consider a relation $R(A, B, C, D, E)$ with functional dependencies:

$$D \rightarrow A \quad B, C \rightarrow D \quad A, B, C \rightarrow D, E \quad C \rightarrow D \quad B, E \rightarrow A, C$$

Compute a canonical set of functional dependencies.

4. We consider a relation $R(A, B, C, D, E)$ with **canonical** set of functional dependencies:

$$B, C, D \rightarrow E \quad A \rightarrow B \quad E \rightarrow B \quad E \rightarrow C \quad A, E \rightarrow D \quad A \rightarrow C$$

Use the synthesis algorithm to compute the Boyce-Codd normal form of R .

5. We consider a relation $R(A, B, C, D, E)$ with **canonical** set of functional dependencies:

$$B, C, D \rightarrow E \quad A \rightarrow B \quad E \rightarrow B \quad E \rightarrow C \quad A, E \rightarrow D \quad A \rightarrow C$$

Use the synthesis algorithm to compute the third normal form of R .

6. Consider the transformation to BCNF and 3NF above. What functional dependencies have been lost in these transformations?